

SOFTWARE REQUIREMENTS SPECIFICATION

DG Fuel Management & Vendor Workflow

Flipkart Minutes - Dark Store Operations

Document Type

Base Scope

Optional Scope

Platform

Software Requirements Specification (SRS)

DG Fuel Management & Vendor Workflow

Vendor Billing & Fusion Integration

IoSense

1. Purpose

- This document describes the requirements for DG fuel monitoring, vendor refueling and DG maintenance/breakdown workflows for Flipkart Minutes stores on the IoSense platform.
- The Base Scope covers fuel monitoring, low-fuel alerts, refueling work orders, vendor refueling, preventive maintenance, breakdown maintenance and operational views.
- The Optional Scope covers vendor billing, automated reports and Fusion integration.
- Any point that is not finalized is marked as TBD instead of being assumed.

2. Scope

2.1 Base Scope

- DG and store setup.
- Vendor and SPOC management.
- DG fuel-level monitoring and low-fuel alerts.
- Automatic work-order creation and assignment.
- Vendor acknowledgement and refueling.
- Vendor SPOC entry of the fuel quantity added.
- Work-order completion after vendor submission.
- Dashboards, transaction views and downloads.
- Role-based access and audit history.
- Maintenance vendor and SPOC management.
- Maintenance work-order creation, assignment, acknowledgement and completion.
- Preventive maintenance and breakdown maintenance workflow for DGs.

2.2 Optional Scope

- Vendor billing and invoice management.
- Automated report generation/distribution.
- Fusion integration and sending invoice to Fusion via API

3. Store and DG Setup

- Each store will normally have one primary DG, Each DG will be identified by their unique DG ID.
- The system shall also allow additional DGs to be added to a store when a mobile, standby or additional DG is placed at the site. Transactions against that DG will be stored separately and will be linked to respective DG ID.
- Stores will be organized under City and Zone.
- DG capacity and other DG details (Fuel Level (%), Estimated Current Fuel Quantity (L), Running Hours, Energy Generation and DG Operating Status) will be maintained against the DG and assigned store.
- There will be separate line items for diesel rate and running hours rate which could be set at DG level, store level, city level or zone level.
- For each DG, preventive maintenance settings will be maintained, including the running-hours limit and/or a fixed time period after which preventive maintenance is due.

- The running-hours limit and fixed maintenance period will be configurable for each DG as applicable.
- When the configured running-hours limit is reached or the configured maintenance period is completed, whichever occurs first, the system will raise a preventive maintenance alert and create a maintenance work order.
- A DG breakdown will also create a maintenance work order immediately.
- DG IoT devices will be onboarded to IoSense for monitoring and dashboarding purposes.

4. Vendor and SPOC Setup

- The system will support two types of vendors for DG operations:
- Refilling Vendor – responsible for DG fuel refilling activities.
- DG AMC / Maintenance Vendor – responsible for preventive maintenance and breakdown-related maintenance activities.

4.1 Refilling Vendor

- Multiple Refilling Vendors can be assigned to a single DG/site for risk mitigation.
- One Refilling Vendor can be identified as the Primary Refilling Vendor, while other mapped vendors can be maintained as Backup/Secondary Vendors.
- Vendorwise refilling charges can be set as per the initial alignment.
- One primary SPOC will be mapped per Refilling Vendor.
- A Refilling Vendor can have multiple SPOCs.
- Refueling work orders will be assigned to the mapped Refilling Vendor and applicable SPOC.

4.2 DG AMC / Maintenance Vendor

- Multiple DG AMC / Maintenance Vendors can be assigned to a single DG/site, where required.
- The applicable DG AMC / Maintenance Vendor will be assigned to preventive maintenance and breakdown maintenance work orders for the mapped DG/site.
- One primary SPOC will be mapped per DG AMC / Maintenance Vendor.
- A DG AMC / Maintenance Vendor can have multiple SPOCs.

4.3 Vendor and SPOC Mapping

- If the assigned vendor or primary SPOC is inactive, the DG will be shown as requiring valid vendor/SPOC mapping for the relevant activity.
- An authorized user can reassign an active work order to another mapped vendor/SPOC of the relevant vendor type.
- The vendor that actually performs the refueling or maintenance activity will remain linked to the respective transaction or work order.
- Historical transactions and work orders will remain available even if a vendor or SPOC is deactivated.

4.4 Same Vendor for Both Activities

- The same vendor can be both a Refilling Vendor and a DG AMC / Maintenance Vendor at the same time.
- In such a case, the vendor can have separate SPOCs or the same SPOC for the two responsibilities, as configured in the system.

5. User Profiles and Role-Based Access

- **Store User** — Has access only to the store(s) assigned to them. They can view DG details, fuel status, alerts, work orders and transactions for their assigned store(s). The Store User can also add additional DGs for their assigned store(s), such as mobile or standby DGs, and map multiple vendors and their SPOCs to the DG/site as required.
- **Central User** — Has access to all stores, dashboards, transactions, work orders and reports. They can filter and download data by Zone, City, Store and Vendor. They can also upload and modify the Store Master, including Zone, City, Store and relevant email IDs.
- **Vendor User / Vendor SPOC** — Has access only to the DG/store(s) assigned to their vendor account. They can view and acknowledge assigned work orders, carry out refueling, manually enter the fuel quantity added, upload supporting proof where applicable, and submit the refueling details against the same work order. They can view/download the transactions available to them, including transactions relevant to optional billing.
- **Super Admin** — There will be one Super Admin user. The Super Admin has access to all stores, vendors, dashboards, reports and system configurations. They can configure the DG low-fuel threshold and the waiting time before a low-fuel alert is raised, and manage system-level settings and user access where required.

6. Fuel Monitoring and Low-Fuel Alert

- The system will use the DG fuel data available in IoSense for the agreed monitoring use case after IoT onboarding.
- Fuel Level (%) will be used to identify a low-fuel condition where applicable.
- Fuel-level limits can be configured at Global, Zone, City, Store or DG level.
- The most specific setting will take priority. For example, a DG-level setting will take priority over a Store-level setting.
- The system can be configured to wait for a specified amount of time before raising a low-fuel alert.
- For example, if the fuel level goes below the configured limit and stays there for the configured time, an alert can be raised. If it comes back above the limit before that time, the alert will not be raised.
- The exact waiting time will be finalized based on the actual DG data frequency and agreed requirement.

7. Low-Fuel Alert and Work Order

- When the low-fuel condition meets the agreed criteria, the system will create one alert and one work order.
- The system will avoid creating duplicate alerts or duplicate work orders for the same ongoing low-fuel condition.
- Each work order will have a unique Work Order / Refill Request ID.
- The same ID will be used for the complete refueling transaction and for the optional billing process.

8. Work Order Assignment

- The system will automatically assign the work order to the active vendor and primary SPOC mapped to that DG.

- If there is no valid active vendor or SPOC, the work order will not be assigned and the DG will be shown as requiring mapping.
- The initial work-order notification will be sent to the relevant vendor/SPOC and internal users according to their access.

9. Workflow 1 – Vendor Refueling

- The vendor SPOC will acknowledge the work order and carry out the refueling.
- After refueling, the vendor SPOC will enter the fuel quantity added manually in litres.
- The vendor SPOC can upload supporting proof if required; proof upload is optional.
- The vendor SPOC will submit the refueling details against the same work order.
- The fuel quantity entered by the vendor SPOC will be recorded as part of the transaction.
- The platform will not calculate the fuel quantity from pre-refill and post-refill IoT readings as part of this workflow.

10. Workflow 2 – DG Preventive Maintenance and Breakdown

- Preventive maintenance can be triggered by either the configured DG running-hours limit or the configured fixed maintenance period.
- When the running-hours limit is reached or the fixed maintenance period is completed, whichever occurs first, the system will raise a preventive maintenance alert and create a maintenance work order.
- A DG breakdown will also create a maintenance work order immediately.
- The maintenance work order will be linked to the relevant DG and store and assigned to the mapped DG AMC / Maintenance Vendor and applicable SPOC.
- The initial work-order notification will be sent to the assigned maintenance vendor/SPOC and relevant internal users according to their access.
- The vendor SPOC will acknowledge the maintenance work order and carry out the required maintenance activity.
- The vendor SPOC will enter the required maintenance details and upload supporting proof where applicable.
- The vendor SPOC will submit the maintenance details against the same maintenance work order.
- Once the vendor SPOC submits the maintenance details, the maintenance work order will be completed as per the agreed workflow.

10.1 Maintenance Work Order Status

- Main flow: Created → Assigned → Acknowledged → Maintenance in Progress → Submitted → Completed.
- The exact status names can be finalized during UI design if required.

10.2 Maintenance and Breakdown Details

- The system will capture the maintenance trigger type, such as Running Hours, Scheduled Period or Breakdown.
- For running-hours maintenance, the relevant running-hours limit and trigger value will be recorded.
- For scheduled maintenance, the maintenance due date/period will be recorded.
- For breakdown maintenance, the breakdown details entered by the applicable user/vendor will be recorded.
- The maintenance work order will retain the relevant vendor/SPOC, DG, store, creation, acknowledgement, maintenance and completion details.

11. Work Order Completion

- Once the vendor SPOC submits the refueling details, the work order will be completed as per the agreed workflow.
- The manually entered fuel quantity will remain linked to the work order and transaction.
- The same Work Order ID will be retained for the complete transaction.

12. Deactivation and Reassignment

- If a DG is made inactive, new work orders will not be created for that DG.
- If a vendor or SPOC is made inactive during an active work order, the work order will remain available but will require a valid mapping of another SPOC or new vendor
- An authorized user can assign another vendor or SPOC to the same work order.
- The same Work Order ID will continue after reassignment.
- Historical work orders and transactions will not be deleted when a DG or Vendor/SPOC is deactivated.

13. Work Order Status

- Main flow: Created → Assigned → Acknowledged → Refueling → Submitted → Completed.
- The exact status names can be finalized during UI design if required.

14. Dashboards and Views

- Users will be able to view DG, store, zone and city information according to their access.
- Dashboards can show fuel status, DG information, alerts and work-order status.
- Users can view and download transactions and work orders for the selected Store, City, or Zone, with an option to filter the data by Vendor. The available data will be limited to the user's assigned access and role.

15. Transaction Download and Access

- Users can download transactions based on their assigned access.
- A user with access to a Zone can download transactions for the selected Zone/City and all stores in that Zone/City.
- A user with access to a Store can download transactions for that Store.
- Users cannot download transactions outside their permitted access.

16. Notifications

- Only the initial work-order/start notification is required.
- The vendor/SPOC and relevant internal users will receive the initial notification according to their access.
- Other status changes can be viewed directly in the platform.
- Separate notifications for every work-order status change are not required.

17. Reports and Audit History

- Users can filter and download operational data based on their access.
- Filters can include Zone, City, Store, DG, Vendor, date range and work-order/refill/maintenance status.

- Automated reports can be provided as an optional feature. The detailed schedule and recipients will be finalized later.
- Data retention will follow the existing IoSense platform data-retention policy.
- The system will maintain the history of important changes, alerts, work-order actions, refueling transactions and user actions.

18. Optional Scope - Vendor Billing

- The vendor will first select the billing period.
- The vendor will then select the billing method (this will be per site as for each site the billing method could be different):
- Run Hours - billing based on the DG run hours.
- Fuel Consumption - billing based on the fuel consumption recorded for the transaction.
- Based on the selected period and billing method, the system will show the transactions that are eligible for billing.
- The vendor will be able to view and download the eligible transactions before preparing the invoice.
- Transactions that were already included in an earlier billing/invoice will automatically be excluded.
- The eligible transaction list will show the relevant billing information for the selected method.
- The vendor will then enter the invoice amount and upload the invoice which will be sent to Fusion via API
- Supporting proof upload will be optional.
- The exact Fusion fields, API details, authentication, codes and payload format will be finalized based on the Fusion interface specification from Flipkart.
- After the invoice is submitted, the list of transactions linked to that invoice will be locked.

19. Key Data Captured

- DG and Store details.
- Vendor and primary SPOC details.
- Work Order / Refill Request ID.
- Work-order creation, acknowledgement, refueling and submission times.
- Fuel quantity entered manually by the vendor SPOC.
- Optional supporting proof.
- Billing period and billing method, when the optional billing scope is used.
- Invoice details, when the optional billing scope is used.
- DG preventive maintenance settings, including running-hours limit and/or fixed maintenance period.
- Maintenance trigger type and relevant trigger value/date.
- Maintenance work-order details, including vendor/SPOC, status and maintenance timestamps.
- Breakdown details, when the maintenance work order is raised due to a DG breakdown.

20. Simple End-to-End Base Flow

- Refilling: DG fuel data → Low-fuel condition → Alert → Refueling work order created → Primary Refilling Vendor/SPOC assigned → Vendor acknowledges → Vendor refuels → Vendor SPOC enters fuel quantity → Vendor submits → Work order completed.
- Maintenance: Running-hours limit reached OR scheduled maintenance period reached, whichever occurs first, OR DG breakdown → Maintenance work order created → AMC/Maintenance Vendor/SPOC assigned → Vendor acknowledges → Maintenance carried out → Vendor submits maintenance details → Maintenance work order completed.

21. Simple Optional Billing Flow

- **Vendor selects billing period → Vendor selects Run Hours or Fuel Consumption → System shows eligible transactions → Previously billed transactions are removed → Vendor views/downloads transactions → Billing set is prepared → Vendor enters invoice amount → Vendor uploads invoice → Fusion submission if enabled.**